Syed Anwar *Senior Data Scientist*

syedanwarafridi00@gmail.com
+923344546885
Islamabad, Pakistan
https://github.com/syedanwarafridi
kaggle https://www.kaggle.com/syedanwarafridi
linkedin.com/in/syed-anwar-715a411aa
syedanwarafridi.github.io

PROFILE

Enthusiastic and results-driven professional with a unique blend of skills in data science.Leveraging a strong foundation in statistical analysis, machine learning, and deep learning, I am dedicated to transforming raw data into actionable insights and driving informed decision-making.

PROFESSIONAL EXPERIENCE

Senior Data Scientist Jul 2025 – Present | Islamabad, Pakistan
Appedology

Lead Data Scientist Feb 2025 – Jul 2025 | Islamabad, Pakistan
Alpha Squad

- **Built and fine-tuned** a persona-driven crypto Twitter agent (*Mind of Pepe*) by scraping and cleaning tweets from the Aixbt agent, and training a LLaMA model to mimic its style—resulting in **98% signal accuracy** and viral tweet engagement.
- **Developed** a gated terminal system that integrates **AI-based signal generation**, **crypto sentiment analysis**, and **technical analysis** using real-time data from **CoinMarketCap**—delivering actionable insights to meme coin traders.
- **Integrated** a **Retrieval-Augmented Generation (RAG)** system with the agent and terminal, enabling contextual understanding and enhancing user relevance by pulling live crypto data and user behavior patterns.
- **Developed** a full AI-powered receptionist system using *n8n*, Telnyx, and ElevenLabs for **Jumper Media**, enabling intelligent call handling, event scheduling, follow-ups via email, and outbound calls.
- **Integrated** real-time **speech-to-text transcription** and Retrieval-Augmented Generation (RAG) with webhook-driven context from Jumper Media's **customer emails and call logs**, enhancing personalization and relevance.
- **Optimized** system performance by reducing response latency to **850ms**, delivering a seamless and human-like user experience across all interaction channels.

Data Scientist Aug 2024 – Feb 2025 | Lahore, Pakistan
MetaViz Pro

I am involved in developing Computer Vision Projects with Deep Learning and Natural Language Processing (NLP) Systems from scratch training of Transformers, Large Language Models (LLM's) and Agents. Also Applying Data Science on Businesses for Solving Business Problems.

- I trained two Transformer-based models from sctach: one for text generation with 5 million parameters, and the other for code completion in Python and Java with 19 million parameters. The models were developed using GPT-NeoX and PyTorch. The text generation model achieved 74% accuracy on the testing data, while the code completion model achieved 67% accuracy.
- Develop a system for real-time yoga pose estimation using computer vision, where extracted pose features are compared with a database of reference poses. Provide feedback to guide users in correcting their posture based on the comparison.
- I developed a system that detects and recognizes workers' faces to mark attendance. Additionally, it monitors for the absence of PPE (Personal Protective Equipment) and triggers an alarm if non-compliance is detected, while also identifying the worker involved.
- I trained a Convolutional Neural Network (CNN) model for a researcher, achieving 98% accuracy on a brain tumor dataset. While previous research has attained similar accuracy using deep neural networks like EfficientNet, I achieved this performance with a more compact architecture, utilizing only 5 layers of CNN.
- I developed a system using AI agents in Langgraph that assesses a user's current job and scores the likelihood of it being replaced by AI in the future. The system also provides recommendations for alternative careers, performs a skills gap analysis, and outlines potential career pathways. Additionally, the system incorporates various other AI-driven features to support users in their career development.
- Managed and cleaned complex financial data, developed machine learning models for insights, and analyzed trends to support strategic financial decisions.

JMM Technologies

Data Scientist Sep 2023 – Aug 2024 | Peshawar, Pakistan

Engaged in developing Computer Vision Projects with Deep Learning and Natural Language Processing (NLP) Chatbots Using Large Language Models (LLM's). Also Applying Data Science on Businesses for Solving Business Problems.

- Developed an AI web app using LLMs to detect errors in translation documents (Spanish, French, Arabic) compared to English source docs, achieving the high accuracy with openAI GPT-4o LLM. Enhanced accuracy from 50% to 90% through advanced prompt engineering. Reduced costs by 40% via token management and Python regex implementation.
- Spearheaded a US company's end-to-end integrated system, deploying advanced Regression Algorithms for precise football score predictions. Expertly cleansed data from the provided API for optimal input quality.
- The developed AI system simplifies creating custom Language Model Models (LLMs) by incorporating various content types like YouTube videos, PDFs, and website data. It utilizes the openAI assistant API, LangChain for YouTube video transcriptions, and web scraping for LLM context, enabling efficient generation of Retrieval-Augmented Generative Systems (RAGS) and Code Interpreters.
- Trained and finetuned Yolov8 and Yolov5 for the American Sign Language Detection(at MVP Stage) System for Saudi Airport Assistance, ensuring the high accuracy of detection and tackling the problem of FPS Lag properly.
- Developed Advance Retrieval Augmented Generation Application using techniques like Query Expansion, Cross-Encoder Re-ranking Algorithm. These advance technique helped Improving the overall result of the RAG Application by giving the highly accurate retrievals.
- Utilized Beautiful Soup, Selenium, and Octoparse for sophisticated web scraping, extracting real-time data from authorized websites for various projects.
- Applied expertise in Time Series Modeling (ARIMA, SARIMA, Facebook Prophet) for forecasting temperature and inflation rate trends.
- Leveraged AWS services (EC2, Lambda, S3, Cloud9) for seamless storage and transformation of data sources, showcasing proficiency in cloud computing solutions.

Artificial Intelligence Engineer Sep 2023 – Dec 2023

- Spearheaded a US company's end-to-end integrated system, deploying advanced Regression Algorithms for precise football score predictions. Expertly cleansed data from the provided API for optimal input quality.
- Crafted impactful sales dashboards with Power BI, transforming intricate datasets into actionable insights for informed decision-making.
- Advanced computer vision by fine-tuning Yolov8 and Yolov5 models for an MVP-stage American Sign Language Detection System, significantly boosting accuracy and eliminating FPS lag at a Saudi airport.
- Developed an innovative LangChain and OpenAI API project for extracting answers from user-uploaded documents, showcasing a fusion of cutting-edge technologies.
- Played a pivotal role in a project for 'Urban E Recycling Company,' applying adept data cleaning and analysis techniques for valuable insights.
- Utilized Beautiful Soup, Selenium, and Octoparse for sophisticated web scraping, extracting real-time data from authorized websites for various projects.
- Applied expertise in Time Series Modeling (ARIMA, SARIMA, Facebook Prophet) for forecasting temperature and inflation rate trends.
- Leveraged AWS services (EC2, Lambda, S3, Cloud9) for seamless storage and transformation of data sources, showcasing proficiency in cloud computing solutions.

Machine Learning Engineer Associate Nov 2022 – May 2023 | Quetta, Pakistan

Government Innovation Lab (United Nation Development Program UNDP)

- Led the development of a Smart Attendance System for the Planning and Development Department, implementing machine learning techniques to enhance accuracy and streamline attendance management.
- Made significant contributions to the Smart Vehicle Counting project, leveraging machine learning algorithms and innovative technologies for precise and automated vehicle counting.
- Demonstrated project leadership, collaborated with cross-functional teams, and integrated cutting-edge technologies to achieve successful outcomes during the Fellowship as an ICT Developer at the Government Innovation Lab.

Data Science Intern Jun 2022 – Jul 2022 | Quetta, Pakistan

British Airways

- Conducted in-depth data analysis at British Airways, identifying patterns and factors contributing to ticket cancellations, leading to targeted strategies for mitigation.

- Led the training of predictive models, utilizing machine learning algorithms to optimize operational performance, enhance insights, and forecast trends.
- Translated complex data findings into actionable insights, communicated recommendations effectively, and played a direct role in improving profit margins and operational efficiency during a virtual internship.

Research Assistant Jan 2021 – Dec 2021 | Quetta, Pakistan
University of Balochistan (Computer Science Department)
Contributed to cutting-edge research as a Research Assistant at the University of Balochistan, executing advanced model training and refining predictive algorithms for projects on Gender Predictions and License Plate Detection.

SKILLS

Programming Languages
Python, C++, JavaScript, MySQL, PostgreSQL

Web Frameworks
HTML, CSS, Django, FastAPI, Flask, Dash, Gradio, Streamlit

Cloud Computing Platforms
PythonAnywhere, Amazon Web Services (AWS), Google Cloud Platform (GCP), IBM Cloud, DigitalOcean, Azure OpenAI, Azure AI Search, Azure Document Intelligence, Azure App Services, Azure Container Apps, Azure Function Apps, AWS S3, AWS EC2, AWS Kinesis

Liberaries & Frameworks/Tools
TensorFlow, Keras, Scikit Learn, PyTorch, Ultralytics, Pillow, OpenCv, Pandas, Numpy, Matplotlib, Seaborn, Plotly, Power Bi, Selenium, BeautifulSoup, LangChain, HuggingFace, Pydantic, OpenAI, Langgraph, NLTK, Git, PySpark, CrewAI, LlamaIndex, Transformers, Langflow, DVC, DBT, Snowflake, MLFlow, Weights & Biases, CI/CD

Hardware
Jetson Nano, Raspberry Pi, Arduino, OpenCV OAK-D

PROJECTS

Transformers From Scratch
Tech Stack: Python, GPT-NEOX, CodeXGLUE, PyTorch
- **Text Generation Model**: Trained a Transformer-based model with 5 million parameters using GPT-NeoX and PyTorch, achieving 74% accuracy in generating contextually relevant text.
- **Code Completion Model**: Developed a Transformer model for Python and Java code completion with 19 million parameters, also built with GPT-NeoX and PyTorch, and achieved 67% accuracy.
- **Architecture and Evaluation**: Both models were fine-tuned on task-specific datasets and evaluated using accuracy, demonstrating strong performance with relatively compact architectures.

ProofLingo
Tech Stack: OpenAI, Langchain
- Developed an AI system to detect contextual, missing, consistency, and incorrect translation errors by comparing translations of Source with Target language documents using GPT-4 LLM, achieving over 90% accuracy.
- Reduced system costs by 40% by managing tokens and implementing prompt engineering.
- Created AI tools for text modification (bold, italic, color changes), bilingual document queries, tone adaptation in translations, and rephrasing with alternative words, and collaborated with the backend team to build RESTful APIs for CRUD operations.

Sign Language Detection
Tech Stack: TensorFlow, Keras, OpenCv, Ultralytics, OAK-D,
- In a Saudi Govt initiative, proficiently trained YOLOv8, LSTM, and CNN1D models for Sign Language Detection.
- Seamlessly integrated these advanced models with ChatGPT, fostering enhanced language processing capabilities.
- Contributed to the project's success by combining computer vision and language understanding technologies for effective Sign Language Detection in collaboration with ChatGPT.

Smart Attendance System
Tech Stack: Scikit-Learn, OpenCv,Numpy, Flask, HTML, CSS, Ajax
- Spearheaded the development of an innovative Facial Recognition-based Attendance System, showcasing expertise in cutting-edge technology.
- Applied advanced training techniques to SVM and KNN models, ensuring precise and reliable recognition of individuals.
- Orchestrated the successful deployment of the system within the Planning and Development Department, resulting in a significant boost in operational efficiency and heightened security measures.

Smart Vehicle Counting System

Tech Stack: OpenCv, LabelImg, JetsonNano

- Actively participated in the end-to-end process of data collection, annotation, and cleaning, laying the foundation for a robust dataset creation.
- Applied expertise to train Yolov8 specifically on vehicle data, elevating the model's proficiency in object detection.
- Played a crucial role in the project's initial stages, ensuring the dataset's quality and optimizing the Yolov8 model for enhanced performance.

Self-Driving Car

Tech Stack: Raspberry Pi, Arduino, OpenCv

- Successfully executed a FYP focused on a Self-Driving Vehicle, integrating cutting-edge IoT and AI technologies.
- Demonstrated proficiency in merging innovative technologies to create an advanced autonomous vehicle system.
- Applied practical skills and theoretical knowledge to contribute to the forefront of self-driving vehicle development.

License Plate Detection

Tech Stack: Python, TensorFlow, OpenCv, Keras,Numpy

- Implemented License Plate Detection for a research paper project under PhD supervision.
- Trained four models (Faster RCNN, RCNN, SSD, Yolov8) to achieve comprehensive license plate recognition.
- Contributed to advancing the field through hands-on application of diverse detection models in a real-world research setting.

EDUCATION

Bachelor's in Computer Science Aug 2018 – Dec 2022 | Quetta, Pakistan

University Of Balochistan

CGPA 3.36/4.00

AWARDS

IEEE Workshop

IEEE

Training program in university from IEEE. in Award I get Certifiticate and also IEEE publication account for free.